

CHAPTER 7: VEHICLE-ACTUATED CONTROL

7.1 INTRODUCTION

- 1 There are a number of different modes whereby traffic signals can be controlled. The most common modes are:
 - (a) Fixed time control in which the sequence and duration of each phase is the same in each cycle (in one timing plan).
 - (b) Vehicle-actuated control in which the appearance and, in the case of vehicular phases, the duration of each phase is dependent upon demands input from vehicle detectors and pedestrian push buttons.
 - (c) Traffic adaptive and responsive control in which the duration of each phase is determined using on-line optimisation models. Traffic adaptive control employs relatively simple models for this purpose, while models that are more complex, are used in traffic responsive control systems.
- 2 Vehicle-actuated signals allow signal indications to vary in accordance with the actuation of vehicle detecting devices. Signal intervals are self-adjustable according to real time traffic demand, within certain limits.
- 3 Vehicle-actuated signals can provide considerable savings in delay when compared to fixed time signals at remotely located (isolated) junctions. However, they tend to increase the number of stops on actuated approaches.

7.2 APPLICATION OF VEHICLE-ACTUATED CONTROL

- 1 Vehicle-actuated control can provide very efficient traffic operations under specific circumstances. Vehicle-actuated control performs best at remotely located, isolated junctions where traffic arrivals tend to be random. Fixed time control is less effective at isolated junctions where it cannot readily adapt to vehicle arrivals.
- 2 Vehicle-actuated control tends to become ineffective in signalised networks where traffic arrivals are cyclic. Efficient traffic control of a network involves not only the control of individual junctions within the network, but also the effective integration of junctions into a co-ordinated system. This will ensure the smooth flow of traffic through the network and minimises stop-and-go cycles. Under these conditions, fixed time control is more suitable than actuated control.
- 3 Vehicle-actuated control can, however, be incorporated into a linked co-ordinated system by providing a common cycle length as a background cycle, so that side-street demands do not interfere with progression along the main street. A vehicle on the side street may only place a demand for a green interval during a predetermined time "window". Any unused green on side-street phases can be transferred to the arterial. Linked vehicle-actuated control can provide benefits on arterial streets, but is less effective in more involved networks.

7.3 STRATEGIES OF VEHICLE-ACTUATED CONTROL

- 1 In vehicle-actuated control, signal timings are varied according to the presence of traffic flows, making optimum use of green times to serve phases only when there is a demand. Different actuated control strategies have been developed, but all with the same objective, **to keep the junction busy and to eliminate idle green time, thus minimising delay to vehicles.**
- 2 One strategy of actuated control is the **volume-density** controller, which has the ability to calculate the duration of green based on actual demand. It also has the feature that intergreen intervals can be adjusted according to actual speeds, thereby reducing lost time during the intergreen.
- 3 The **basic vehicle-actuated control strategy** is commonly used in South Africa. The objective of this strategy is to improve operations with a control strategy which is simple and robust to implement. The lengths of the intergreen intervals are established as for fixed time control and only the green periods are varied.
- 4 The basic vehicle-actuated control strategy attempts to minimise delay by extending green phase only for the duration queues are present on approaches receiving green. A green period that has been terminated before a queue has been served will result in a queue being cut off too early. The overflow of vehicles can only be discharged in subsequent greens, which increases delay. If the next green phase is again too short, the queue will grow, causing excessive delay (the signals are oversaturated).
- 5 However, should a too long green period be provided, it would result in lost efficiency every time that the departure flow rate drops below the maximum saturation flow rate. Such maximum saturation flow rate is normally only achievable when there is a queue of vehicles on an approach. The delay on approaches receiving a red signal increases while approaches receiving green light signals are operating at low levels of efficiency.
- 6 The method used to detect the end of a discharging queue is by searching for a gap between two vehicles. A long gap would normally only appear after a queue has discharged, thus indicating that the queue has departed. A gap is found if no further vehicles are detected over a certain distance, called the detection area, as illustrated in Figure 7.1.

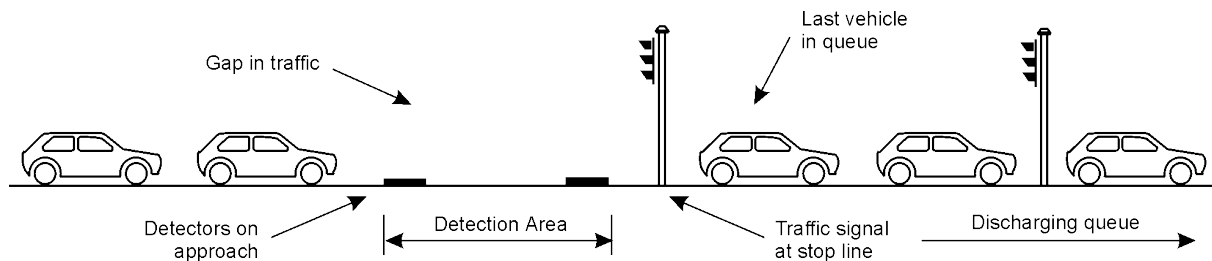


Figure 7.1: Searching for gaps in the traffic stream

7.4 FULLY- AND SEMI-ACTUATED CONTROL

7.4.1 General

- 1 Detectors on all the lanes or approaches to a junction are not always needed. If detectors are used only on some lanes or approaches, it is referred to as semi-actuated control. In fully-actuated control, detectors would normally be used on all lanes or approaches.
- 2 Fully-actuated control allows for the extension of all green intervals provided at a signal. This mode of control finds primary application at isolated junctions since it cannot be co-ordinated with other signals without losing the flexibility for which it was designed.
- 3 Semi-actuated control is used to call and extend only a subset of phases at a traffic signal, such as a right-turn or the side-street phase. Semi-actuated control can also be used in a linked or co-ordinated signal system.

7.4.2 Fully-actuated control

- 1 *Fully-actuated control* allows for the extension of all green phases provided at a traffic signal.
- 2 In a fully-actuated signal, the controller displays a particular signal stage, usually on the main road, for a minimum period of time (minimum green), after which it "rests" in this stage until a demand for a right of way change is detected.
- 3 The maximum green timer is started at the time a demand for a right of way change is detected. If the current minimum green timer has not elapsed, the controller waits for minimum green to terminate. The controller then waits for a gap in the traffic, subject to the maximum green interval timer.
- 4 Once a gap is detected or the maximum green expires, the signals are changed to a new stage by the controller (a "gap change" occurs if this is due to a gap in the traffic stream). The full logic is repeated for the next signal stage.

7.4.3 Semi-actuated control (isolated)

- 1 *Semi-actuated control* can be used at isolated junctions or in linked co-ordinated signals systems. At isolated junctions, it would normally operate with a variable cycle length, and there is no need to transfer any time saved during one stage to another stage.
- 2 In such a system, the main street is usually favoured and the main street green phase is always provided. The main street receives a minimum duration of green, after which it "rests" in this green until a demand for a right of way change is detected on either a right-turn lane or on a side street.
- 3 The controller only switches to green after a minimum time period, to allow for the accumulation of a number of vehicles on stopped approaches before a green light signal is provided.
- 4 The duration of the right-turn or side-street green is either fixed, or extendible up to a maximum interval. Once a gap is detected, or maximum green has expired, right of way reverts back to the major street, and the operations are repeated.

7.4.4 Semi-actuated control (linked)

- 1 *Linked semi-actuated control* is used in co-ordinated signal systems in which a fixed cycle length must be maintained, while a window of minimum green must be guaranteed for the main road.
- 2 This can be achieved by allowing the signal to operate according to a fixed time plan. At the termination of a particular signal stage, demands for other stages are examined.
- 3 If no such demand was detected, stages can be skipped. If a demand was detected, a green interval is provided which is either of fixed duration or extendible between minimum and maximum limits.
- 4 Any unused green time is allocated to other stages as specified (usually the main stage), whether or not it is actually required by such a stage.
- 5 In allocating spare green time to other stages, care must be taken to ensure that the window of minimum green is maintained on the main road.

7.5 MINIMUM AND MAXIMUM GREEN

7.5.1 Minimum green

- 1 Minimum green is the minimum time the green light signal must be displayed, measured from the instant it appears, irrespective of the demand for a right of way change.
- 2 The minimum green is provided to allow for the following:
 - (a) Absolute minimum green times required for safety (refer to Chapter 3 of this manual).
 - (b) If pedestrian signals are provided, sufficient time for pedestrians to enter and clear the junction (refer to Chapter 4 of this manual).
 - (c) Clearance of vehicles waiting between the vehicle detectors and the stop line. The tendency in most installations is not to leave any space between the detectors and the stop line (at least not longer than 1 or 2 vehicles that can be accommodated by the safety minimum green). Where such space is provided, the minimum green would depend on the number of vehicles waiting, starting lost time and saturation flow.

7.5.2 Maximum Green

- 1 Maximum green is used to terminate a running green light signal irrespective of the extension period in order to prevent vehicles receiving red light signals from waiting indefinitely because of a continuous stream of traffic on the running green signal. Timing of the maximum green interval commences as soon as a demand is recorded for a right of way change.
- 2 Under low and medium traffic flow conditions, a maximum green interval of between 30 to 60 seconds (or approximately 40 seconds) is normally provided at fully-actuated controlled signals.
- 3 Under heavy traffic volumes, for example during peak periods, the green period may run successively to maximum, giving in effect, fixed time operation. Under such conditions, the value of maximum green may therefore be determined as for fixed time signal control (refer to Chapter 6 of this manual).

7.6 STAGE REVERSION

- 1 When no demand is detected for a signal phase, the traffic signal should revert to a nominated stage and "rest" in this stage. The following options are available:
 - (a) **Rest in main green.** The signal rests in the main signal phase. No need exists to detect the presence of vehicles on the approaches served by this phase and only extension detectors may be required.
 - (b) **Rest in the last requested green (nil revert).** The controller rests in the last green in the absence of demand. All approaches must be equipped with calling detectors.
 - (c) **Rest in all-red.** The signal reverts back to red light signals on all approaches in the absence of demands. Right of way is provided on a "first come - first serve" principle.

- 2 The advantage of the "rest in all-red" method is that the lost time due to the intergreen period is reduced. The controller can quickly switch to green when a vehicle is detected on any approach, which allows the approaching vehicle to drive through the junction without having to stop. There is, however, the danger that this may result in unsafe driver behaviour since drivers may become accustomed to a green signal when arriving at the junction.
- 3 The "rest in the main green" has the advantage that green is provided on the street with the highest traffic volume. This is often the preferred method of control, but care should be taken that a series of green signals are not displayed along a road, which could encourage speeding.

7.7 VEHICLE DETECTION

7.7.1 Detection devices

- 1 A wide range of detecting devices have been developed for use in vehicle-actuated control. The inductive loop detector is currently most widely used in modern traffic control systems. Details of inductive loop detectors are given in Chapter 20 of this manual (Volume 3).
- 2 Inductive loops may operate in one of two modes, namely passage and presence modes. Presence detectors are used to indicate the presence of a vehicle on the detector, while passage detectors are used to indicate that a vehicle has crossed a loop and no indication is given of the time the vehicle has spent on the loop. Passage detectors are normally used for counting traffic while presence detectors are used for the detection of vehicles at vehicle-actuated traffic signals.
- 3 Two sets of detectors are used at vehicle-actuated controlled junctions, namely **extension** and **calling** detectors, although detectors can be used to serve both functions. Extension detectors are primarily used to extend a green period, while calling detectors are primarily used to detect the presence of vehicles waiting at red light signals.

7.7.2 Extension detectors

- 1 Extension detectors are used to extend a green period until a gap is found in the traffic stream. The extension detectors, however, can also be used to register the presence of vehicles waiting at red light signals.
- 2 A variety of extension detectors have been used in the past, ranging from single detectors to so-called large area detection configurations.
- 3 The ideal (but unpractical) method of extending green is by using a single large detector on the approach to a signal. Green is extended for the length of time a moving vehicle remains in the detection area (subject to maximum limits).
- 4 In practice, it is difficult to provide large area detection. Long loops tend not to have adequate sensitivity, with the result that the detectors can miss vehicles. The loops also require more maintenance, and are especially vulnerable during road works.

- 5 Large area detection can, however, be emulated by installing a multiple of short loops. *Extension intervals* are provided which allow vehicles to travel from loop to loop without green being terminated. The extensions are designed for a vehicle travelling at a low speed, but not necessarily for the slowest vehicle. This has the advantage that a vehicle travelling at a very slow speed will not extend the green for an unreasonable long time.
- 6 Although multiple loop detectors are used, these detectors can all be connected to one electronic circuit to extend the same green interval (subject to a maximum limit that can be accommodated by the electronic circuit).
- 7 Extension detectors should preferably have a diamond or parallelogram shape. These shapes are more effective in detecting fast moving vehicles. Rectangular loops can also be used for the purpose of extending green, but are less effective when vehicles are travelling at higher speeds.
- 6 The problem of vehicles stopping before or beyond the detector can also be addressed by using a detector over a longer distance, operating in **non-latching** mode. Detectors of up to 4 m or longer may be required for this purpose. The exact size and position of the detectors should be determined by observing the positions where vehicles typically stop at a junction. The size and position should be such that few vehicles would stop before or beyond the detector or detectors. Speed is of no concern, and rectangular shaped detectors can therefore be used for detecting stopped vehicles.
- 7 A further problem with operating detectors in latching mode occurs on undivided approaches where encroachment of right-turning vehicles from opposing approaches could result in the placement of false calls. This problem can be reduced by using skewed detector loops. Alternatively, a **delay** of a few seconds can also be introduced before the presence of a vehicle is actually registered.

7.7.3 Calling detectors

- 1 Calling detectors are required to detect the presence of vehicles waiting at red signals. These detectors are normally located at or near the *stop line* to reduce the possibility that vehicles become trapped on a red signal when they stop beyond the detector.
- 2 The stop line calling detector can also serve as an extension detector. This, however, is not very efficient since a green interval is extended after the vehicle has already entered the junction. When separate extension detectors are provided at a junction, the stop line calling detectors should NOT be used to extend the green interval.
- 3 Where calling detectors are used to place a call for the same green interval, all such detectors can be connected to one electronic circuit (subject to a maximum limit that can be accommodated by the electronic circuit).
- 4 Care should also be taken to place the calling detector at a position where most vehicles would stop on the detector to reduce the probability of vehicles either stopping before or beyond the detector. Such vehicles will not be detected and could wait for a long time for a green signal.
- 5 The problem of vehicles becoming trapped beyond the calling detector can be addressed by placing the detector in **latching** (or locking) mode. In this mode, the call of the vehicle is held after it has passed the detector. The loops normally have a parallelogram (or diamond) shape to detect vehicles crossing at high speed. This method has the disadvantage that when a vehicle passes without stopping, it would place a false call for a stage that is not required.

7.8 DETECTOR CONFIGURATIONS

7.8.1 General

- 1 A number of example detector configurations that can be used at vehicle-actuated signals are shown in Figures 7.2 to 7.5.
- 2 The configuration given in Figure 7.2 can be used when only a right-turn phase is actuated, while those shown in Figures 7.3 to 7.5 can be used when different types of phases are actuated, such as left-turn, right-turn and main signal phases.
- 3 Care should be taken when applying the examples. Although the examples could probably be used in many applications without modification, there may be specific conditions that would necessitate deviations from the configurations.

7.8.2 Detector configuration for protected/permitted right-turn phases

- 1 The detector configuration shown in Figure 7.2 can be used when a **protected/permitted right-turn phase** is actuated and no actuation is provided for any other phases on the approach. Such a configuration would be operating in a semi-actuated mode of control.
- 2 Only one detector is used on the right-turn lane as both a calling and extension detector. Two possible locations are shown for the detector, depending on whether a leading or lagging phase is provided.
- 3 The detector positions shown in Figure 7.2 have been determined based on the assumption that one or two vehicles can turn right during the main-phase intergreen period. The presence of a second or third waiting vehicle would be the warrant for the separate right-turn phase.

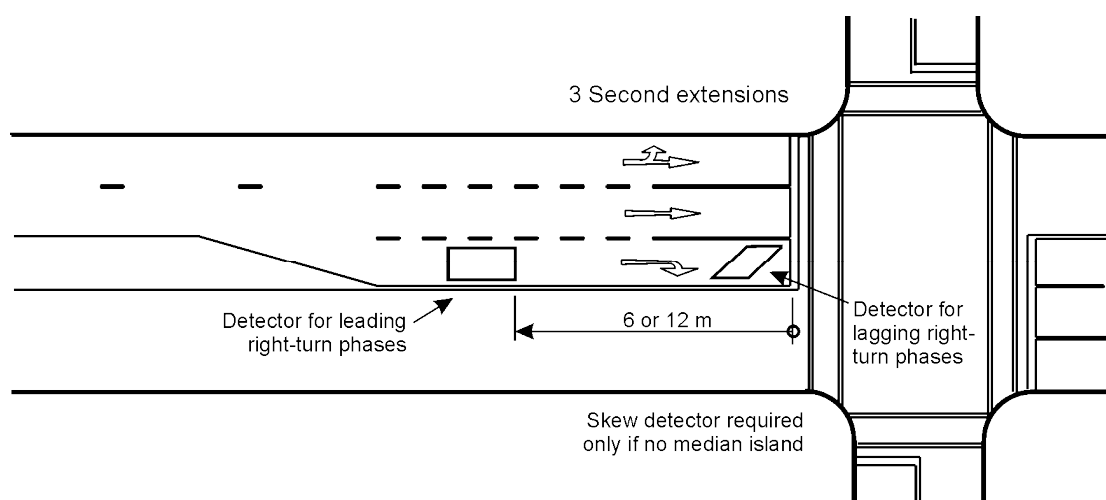


Figure 7.2: Right-turn detector configuration

- 4 Leading right-turn phases require the following detector configuration:
 - (a) The detector should be placed some distance before the stop line since vehicles are stopped on the approach to the junction. A distance of about 6 m can be provided for each vehicle to be skipped (6 m for one vehicle and 12 m for two vehicles). A shorter distance is used at narrower junctions and a longer distance at wider junctions.
 - (b) A detector with a length of about 4 m is generally recommended. Openings between stopped vehicles are seldom more than 4 m and a detector of such length is required to reduce the possibility of vehicles not being detected.
- 5 Lagging right-turn phases require the following detector configuration:
 - (a) Detectors should be provided at or near the stop line since it can be assumed that the first one or two right-turn vehicles are waiting within the junction and not on the approach to the junction. The design of the right-turn detector is then similar to that of other stop line detectors.
 - (b) A delayed-call detector is sometimes used to place a call only if a vehicle is continuously detected beyond a pre-set period of, for example, 5 seconds. This allows the controller to ignore moving vehicles and to accept calls only from stopped vehicles. Care should, however, be taken when using this facility since the fact that a vehicle is moving is not necessarily an indication that there is no longer a right-turn queue.
- 6 The assumptions made in determining the detector positions shown in Figure 7.2 are not valid (and the configuration can therefore not be used) under the following circumstances:
 - (a) When a protected-only right-turn phase is provided.
 - (b) In fully-actuated systems where provision is made for the skipping of the main signal phase.

In both these cases it can not be assumed that the first one or two vehicles will be able to turn right during the main phase intergreen, and stop line calling detectors must be provided to detect the presence of these vehicle.

7.8.3 Single stop line detector configuration

- 1 The single stop line detector configuration shown in Figure 7.3 can be used for the actuation of different types of signal phases (main, left-turn and/or right-turn). This configuration is the simplest available and use is made of only one set of detectors at the stop line. These detectors serve as both calling and extension detectors.
- 2 The disadvantage of this configuration is that green intervals are extended when vehicles have already entered the junction. Each time a gap change occurs, 3 seconds of green may be lost in **addition** to the duration of the intergreen period. This reduces the efficiency of the configuration.

7.8.4 Double detector configuration

- 1 A more efficient configuration is shown in Figure 7.4 in which two sets of detectors are provided, the one used for calling purposes and the other for extension purposes. The extension detectors are set back from the stop line with the result that green extensions are given while a vehicle is still approaching the junction, and not within the junction itself.
- 2 The configuration shown in Figure 7.4 is designed for a speed of 40 km/h. When a vehicle exceeds this speed, some green time will still be lost (although not substantial). A vehicle travelling at 60 km/h only requires about 2 seconds to travel the distance between the extension detectors and the stop line, while a 3 second extension interval is provided. The 1 second lost time is a significant improvement over the lost time of 3 seconds of the single detection system.
- 3 A problem with the configuration in Figure 7.4 is the use of the high design speed of 40 km/h. Many queues of vehicles only depart at a speed of between 20 and 30 km/h, particularly when turning at a junction. A vehicle travelling at a speed of 20 km/h would require approximately 6 seconds to reach the stop line while green is extended for only 3 seconds. It is therefore possible that this vehicle would stop on yellow and not utilise the green extension. This problem can be addressed by the triple detector configuration.

7.8.5 Triple detector configuration

- 1 The triple detector configuration utilises one set of calling and two sets of extension detectors as shown in Figure 7.5. The advantage of the triple detector configuration is that vehicles can be better tracked as they travel over the detection area.
- 2 The configuration shown in Figure 7.5 is designed for a speed of 20 km/h between the first two sets of extension detectors and 30 km/h between the extension detectors and the stop line.
- 3 The triple detector configuration has a slightly lower lost time than the double detector configuration, but the improvement is not significant. The advantage of the triple detector configuration is that it accommodates slower vehicles better than the double detector configuration. A slow vehicle travelling at 20 km/h will be able to reach the second extension detector with the 2 second extension given by the first extension detector. The same vehicle requires slightly more than 3 seconds to reach the stop line, and will therefore arrive at the stop line 1 second after the onset of yellow. Most drivers would not stop at the junction under such circumstances.

7.9 PEDESTRIAN DETECTION

- 1 When designing vehicle-actuated traffic signals, it is important that not only the needs of vehicular traffic are taken into account, but also those of pedestrians (and pedal cyclists).
- 2 When provision is made at a signal to skip a vehicular phase that operates parallel with a pedestrian phase, push buttons should be provided for the actuation of the phase. A call for the phase can be placed by the pedestrian, even if no vehicles are then present to actuate the phase.

7.10 BIBLIOGRAPHY

- 1 DOE (Department of the Environment, UK), 1973, Siting of inductive loops for vehicle detecting equipment at permanent road traffic signal installations, Specification MCE 0108A.
- 2 Kruger, P., 1990, The Efficient Control of an Isolated Vehicle-Actuated Controlled Intersection: A comparison between Queue Control and Volume-Density Control, Institute of Transportation Studies, Report UCB-ITS-RR-90-1, Berkeley, pp.1-66.
- 3 Webster, F.V. and Cobbe, B.M., 1966, Traffic Signals, Road Research Technical Paper No. 56, London, Road Research Laboratory.

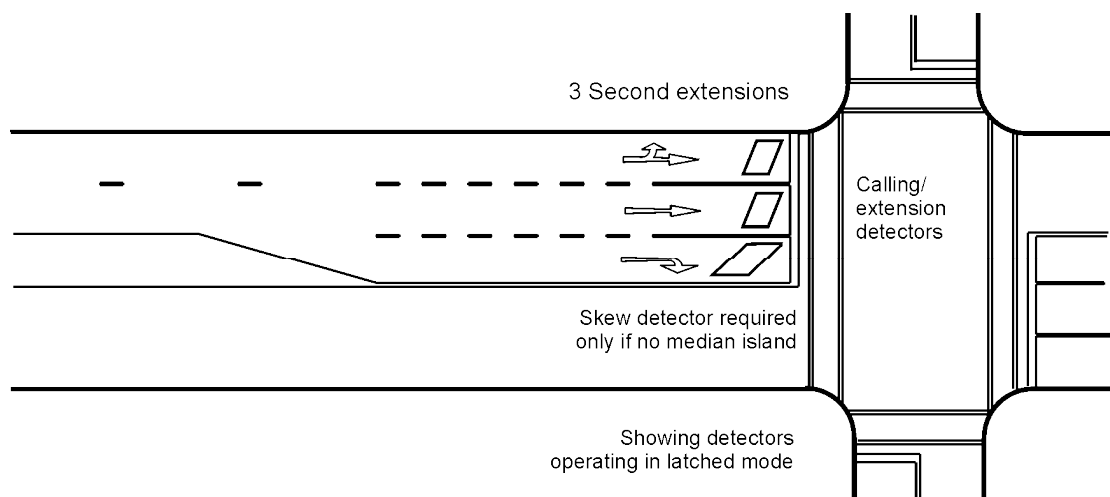


Figure 7.3: Single stop line detector configuration

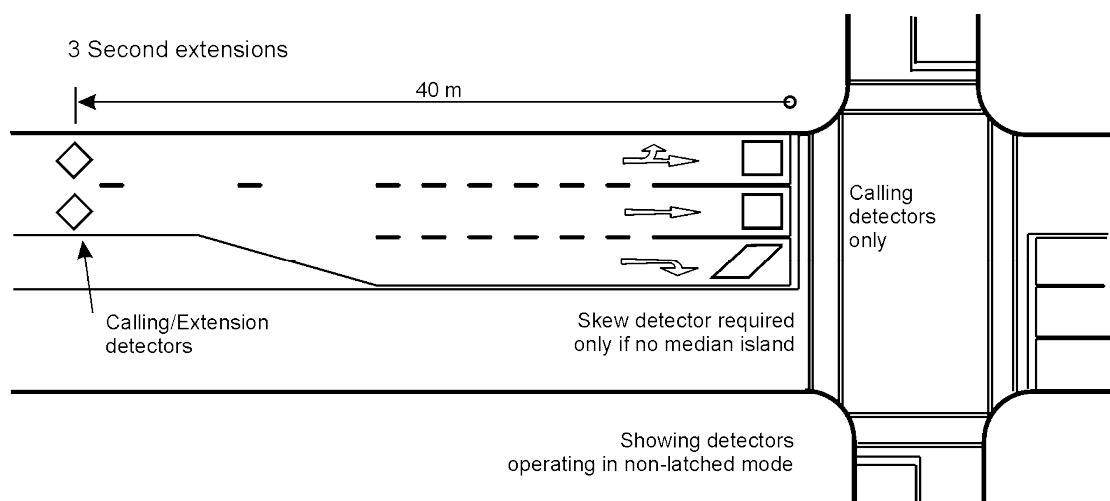


Figure 7.4: Double detector configuration (design speed 40 km/h)

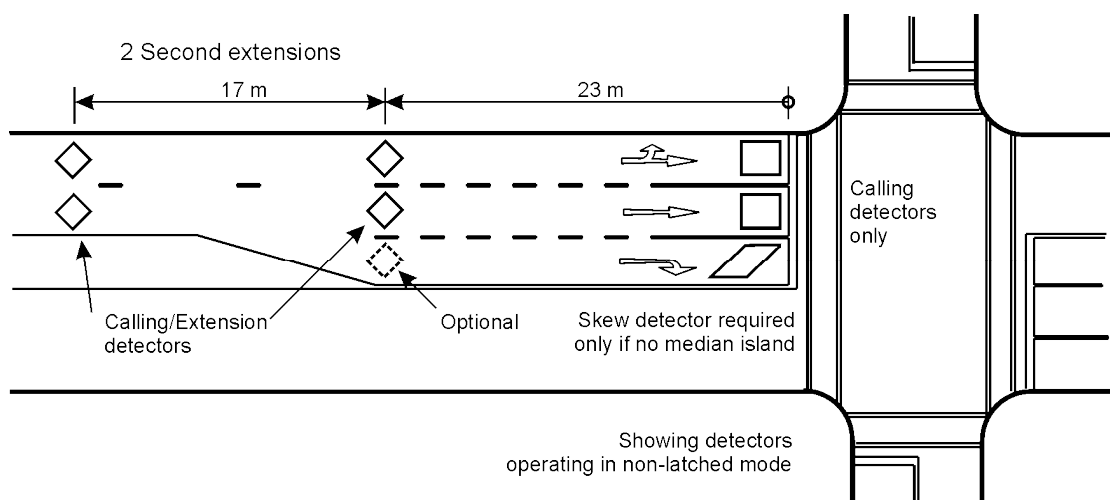


Figure 7.5: Triple detector configuration (design speed 20/30 km/h)

